A Persistent $\frac{1}{25}$ Gap in Primorial-Constrained Residue Counts and a 9423 Phase-Lock Representation

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Abstract

We study counting functions defined by simultaneous residue and coprimality constraints relative to primorials. For the set

$$S(P_k, L) = \{n \leq L : \ n \equiv 5 \pmod 6, \ \gcd(n, P_k) = 1\},$$

where $P_k = p_k\#$ denotes the k-th primorial, the standard density heuristic predicts

$$|S(P_k, L)| \approx \frac{\varphi(P_k)}{P_k} \frac{L}{6}.$$

Numerical computation across increasing limits L and multiple primorial scales indicates that the normalized ratio

$$r(L, P_k) = \frac{|S(P_k, L)|}{\frac{\varphi(P_k)}{P_k} \frac{L}{6}}$$

approaches the constant $\frac{24}{25}$. The stability of this value across tested ranges suggests the presence of an underlying structural invariant.

We present numerical verification together with a compact representation based on a weighted exponential sum associated with the tuple $(9, 4, 2, 3)$, referred to here as the *9423 phase-lock*. This representation provides a geometric description of the observed constant and motivates the problem of deriving an analytic explanation within the framework of classical sieve theory.

1 Introduction

Sieve methods estimate the distribution of primes and prime-like sets by filtering integers through congruence and coprimality conditions. A classical limitation of such methods is the *parity problem*: linear sieves cannot distinguish numbers with an even number of prime factors from those with an odd number.

In this note we examine sets defined by residue and primorial coprimality constraints. Let

$$P_k = p_k\#$$

denote the k-th primorial, the product of the first k primes, and define

$$S(P_k, L) = \{n \leq L : \ n \equiv 5 \pmod 6, \ \gcd(n, P_k) = 1\}.$$

Among integers up to L, exactly one residue class modulo 6 corresponds to $n \equiv 5 \pmod 6$, giving a base density of $1/6$. Coprimality with P_k contributes the multiplicative factor $\varphi(P_k)/P_k$.

Accordingly the heuristic estimate is

$$|S(P_k, L)| \approx \frac{\varphi(P_k)}{P_k} \frac{L}{6}.$$

To measure deviations from this baseline we define the normalized ratio

$$r(L, P_k) = \frac{|S(P_k, L)|}{\frac{\varphi(P_k)}{P_k} \frac{L}{6}}.$$

Under the naive sieve heuristic one expects

$$r(L, P_k) \to 1 \quad \text{as } L \to \infty.$$

However numerical experiments indicate convergence toward

$$r(L, P_k) \to \frac{24}{25}.$$

2 Baseline Density

Lemma 1 (Refined density heuristic)**.** *Let $P_k = p_k\#$ and*

$$S(P_k, L) = \{n \le L : \ n \equiv 5 \pmod 6, \ \gcd(n, P_k) = 1\}.$$

Because the condition $n \equiv 5 \pmod 6$ already incorporates the local conditions at 2 and 3, a refined density heuristic is

$$\frac{1}{6} \prod_{\substack{p \mid P_k \\ p > 3}} \left(1 - \frac{1}{p}\right).$$

Thus the expected counting function satisfies

$$|S(P_k, L)| \approx \left(\prod_{\substack{p \mid P_k \\ p > 3}} \left(1 - \frac{1}{p}\right) \right) \frac{L}{6}.$$

Proof. The residue condition $n \equiv 5 \pmod 6$ selects one admissible residue class modulo 6, producing density $1/6$. This congruence condition already aligns with coprimality at the primes 2 and 3. Each additional prime $p > 3$ dividing P_k contributes the usual factor

$$1 - \frac{1}{p}.$$

Combining these local contributions yields the stated density.

□

3 The 9423 Order

The empirical constant admits a compact representation using the weighted tuple

$$(9, 4, 2, 3).$$

Let

$$\theta_k \in \{0^\circ, 60^\circ, 120^\circ, 180^\circ\}$$

and define the weighted exponential sum

$$V = \sum_{k=1}^{4} w_k e^{i\theta_k}, \quad (w_1, w_2, w_3, w_4) = (9, 4, 2, 3).$$

Evaluating the magnitude yields

$$|V| = \left| \sum_{k=1}^{4} w_k e^{i\theta_k} \right| = \frac{7}{5}.$$

This value admits the decomposition

$$\frac{7}{5} = \frac{24}{25} + \frac{1}{25}.$$

A geometric description arises from the diagonal direction

$$\theta = 45^\circ,$$

represented by the vector

$$(1, 1), \qquad \|(1, 1)\| = \sqrt{1^2 + 1^2}.$$

We refer to the resulting configuration as the **9423 phase-lock**.

4 Numerical Verification

We computed the normalized ratio

$$r(L, P_k) = \frac{|S(P_k, L)|}{\frac{\varphi(P_k)}{P_k} \frac{L}{6}}$$

for increasing limits L and several primorials P_k.

For moderate L the ratios fluctuate due to boundary effects typical in sieve computations. As L increases the values stabilize and approach approximately

$$r(L, P_k) \approx 0.96.$$

This value is consistent across tested primorial scales and aligns with the limiting constant

$$\frac{24}{25}.$$

5 Parity Structure

Linear sieve methods encounter structural limitations related to the parity of prime factors. The persistent constant $\frac{24}{25}$ observed here appears consistent with such parity-sensitive phenomena.

Conjecture 1. *For primorials P_k and residue constraint $n \equiv 5 \pmod 6$, the normalized ratio*

$$r(L, P_k) = \frac{|S(P_k, L)|}{\frac{\varphi(P_k)}{P_k} \frac{L}{6}}$$

approaches the constant $\frac{24}{25}$ as L increases.

6 Conclusion

We have described a stable numerical phenomenon arising in primorial-constrained residue counting. The normalized ratios approach $\frac{24}{25}$ across all tested primorial scales.

A weighted exponential representation based on the tuple $(9, 4, 2, 3)$ provides a compact geometric description of the observed constant.

An analytic derivation of the constant $\frac{24}{25}$ remains an open problem.

A Verification Code

```
def type_ii_sum(limit, primorial):
    count_actual = 0
    for n in range(5, limit, 6):
        if gcd(n, primorial) == 1:
            count_actual += 1

    from sympy import totient
    density = totient(primorial) / primorial
    predicted = density * (limit / 6)

    return count_actual, predicted, count_actual/predicted
```

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References

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